



Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT

JEE-MAIN

Date: 02-01-2026

Time: 09.00Am to 12.00Pm

GTM-08

Max. Marks: 300

KEY SHEET

MATHEMATICS

1	1	2	4	3	3	4	2	5	2
6	1	7	4	8	3	9	4	10	3
11	4	12	2	13	1	14	1	15	3
16	2	17	1	18	3	19	3	20	3
21	2	22	7	23	16	24	4	25	3

PHYSICS

26	3	27	4	28	4	29	2	30	2
31	3	32	4	33	1	34	4	35	2
36	1	37	4	38	4	39	4	40	2
41	1	42	1	43	2	44	1	45	2
46	1	47	25	48	22	49	2	50	5

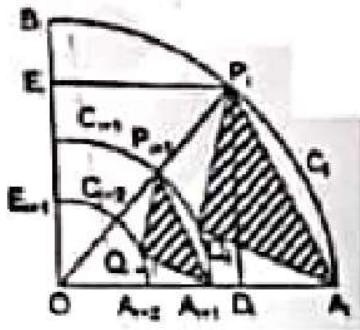
CHEMISTRY

51	3	52	1	53	3	54	3	55	1
56	4	57	3	58	3	59	1	60	1
61	4	62	1	63	4	64	1	65	4
66	4	67	3	68	4	69	3	70	1
71	6	72	100	73	2350	74	4	75	121



SOLUTION MATHEMATICS

1. Sol.



$$A_1 P_1 = \frac{2}{\sqrt{5}} r_1 \text{ and } r_{1+1} = \frac{r_1}{\sqrt{5}}$$

$$\text{And } S_1 = \frac{r_1^2}{5} \rightarrow S_{1+1} = \frac{r_{1+1}^2}{5} = \frac{r_1^2}{25}$$

$$\Rightarrow \frac{S_{1+1}}{S_1} = \frac{1}{5}$$

$$\Rightarrow \lim_{n \rightarrow \infty} (S_1 + S_2 + S_3 + \dots + S_n) = \lim_{n \rightarrow \infty} \left(\frac{\frac{1}{5}}{1 - \frac{1}{5}} \right) = \frac{1}{4}$$

2. $x^2 + 20x - 2020 = 0$ has two roots $a, b \in R$

$x^2 - 20x + 2020 = 0$ has two roots $c, d \in$

Complex

$$ac(a-c) + ad(a-d) + bc(b-c) + bd(b-d)$$

$$= a^2c - ac^2 + a^2d - ad^2 + b^2c - bc^2 + b^2d - bd^2$$

$$= a^2(c+d) + b^2(c+d) - c^2(a+b) - d^2(a+b)$$

$$= (c+d)(a^2 + b^2) - (a+b)(c^2 + d^2)$$



$$= (c+d)((a+b)^2 - 2ab) - (a+b)((c+d)^2 - 2cd)$$

$$= 20[(20)^2 + 4040] + 20[(20)^2 - 4040]$$

$$= 20[(20)^2 + 4040 + (20)^2 - 4040]$$

$$= 20 \times 800 = 16000$$

$$3. \quad I_n = \int_0^\infty x^n e^{-x^2} dx = \int_0^\infty x^{n-1} x e^{-x^2} dx$$

$$\Rightarrow I_n = \left[x^{n-1} \int x e^{-x^2} dx \right]_0^\infty - \int_0^\infty \left\{ \frac{d}{dx} (x^{n-1}) \int x e^{-x^2} dx \right\} dx$$

$$-x^2 = t \Rightarrow I_n = - \left[\frac{x^{n-1}}{2e^{x^2}} \right]_0^\infty + \frac{n-1}{2} \int_0^\infty x^{n-2} e^{-x^2} dx$$

$$\Rightarrow I_n = \frac{n-1}{2} I_{n-2}$$

$$\text{Case I: } n = 2m \Rightarrow I_1 = \frac{(2m-1)(2m-3)(2m-5)\dots 1}{2^n} I_0 = \frac{(2m-1)!}{2^{2m-1}(m-1)!}$$

$$4 \quad p \neq q \neq r \neq 0$$

$$\text{Also, } 2q = (p+r) \quad a+b = -\frac{q}{p}; ab = \frac{r}{p}$$

$$\text{Now, } (a+b)^2 = (a+b) \frac{((a+b)^2 - 2ab)}{ab}$$

$$\text{As, } \Rightarrow ab(a+b) = (a+b)^2 - 2ab$$

$$\therefore \text{ using (1) and (2) in (3), we get } p^2 - 4pr + 3r^2 = 0$$

$$\Rightarrow \left(\frac{p}{r} \right)^2 - 4 \left(\frac{p}{r} \right) + 3 = 0$$



$$\therefore \frac{p}{r} = 3 \left(\text{As, } \frac{p}{r} = 1 \text{ (reject)} \right)$$

$$5. \quad \left| (A^{-1} - I)^2 \right| = \left| A^{-2} - 2A^{-1} + I \right| = \frac{|A^{-1} - 2I + A|}{|A|}$$

$$\therefore A + A^{-1} = \frac{1}{7} \begin{bmatrix} 12 & 25 & 21 \\ 20 & 23 & 77 \\ -14 & 21 & 21 \end{bmatrix} \text{ and } |A| = -7$$

$$\therefore |(A^{-1} - I)^2| = \frac{\begin{vmatrix} -\frac{2}{7} & \frac{25}{7} & 3 \\ \frac{20}{7} & \frac{9}{7} & 11 \\ -2 & 3 & 1 \end{vmatrix}}{-7} = \frac{324}{49}$$

$$\therefore |(A^{-1} - I)^2| = \frac{18}{7}$$

$$\Rightarrow \text{It positive square root } \frac{18}{7}$$

$$6. \quad a_1 = 1, a_2 = 0, a_3 = 1,$$

$$a_n = a_{n-2} + a_{n-3}, n > 3$$

$$a_4 = 1, a_5 = 1, a_6 = 2, a_7 = 2$$

$$a_8 = 3, a_9 = 4, a_{10} = 5, a_{11} = 7$$

$$a_{12} = 9, a_{13} = 12, a_{14} = 16$$

7. Check Continuity (b) Both $y = x^2$ and $y = 4|x| - x^2$ are continuous functions everywhere. The maximum of two continuous functions is always continuous. $\therefore b = 0$ (No points of discontinuity).

Check Differentiability (a) Non-differentiability occurs at sharp corners, typically where the component graphs intersect or where modulus $|x|$ turns.

- Intersection Points: Solve $x^2 = 4|x| - x^2 \Rightarrow 2x^2 = 4|x| \Rightarrow x^2 = 2|x|$. Points are $x = 0, 2, -2$.



- Check Slopes at Critical Points:
- At $x = 0$: Due to $4|x|$, the graph has a sharp "V" shape. (Slope jumps from -4 to 4). $\rightarrow$ Non-differentiable.
- At $x = 2$:
- Slope of $x^2 \rightarrow 2x = 4$.
- Slope of $4x - x^2 \rightarrow 4 - 2x = 0$.
- $4 \neq 0$, so a sharp corner exists. $\rightarrow$ Non-differentiable.
- At $x = -2$: By symmetry (even function), it is also a sharp corner. $\rightarrow$ Non-differentiable.
 $\therefore a = 3$ (Points: $-2, 0, 2$).

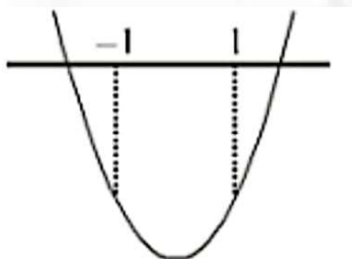
3 Final Answer

$$a + b = 3 + 0 = 3$$

8. $x^2 - 1 = 0 \rightarrow x = \pm 1$

$$y = f(x) = x^2 + (3a - a^2)x - 3a^3$$

$$f(-1) < 0 \Rightarrow 1 - 3a + a^2 - 3a^3 < 0$$



$$\Rightarrow 3a(a^2 + 1) - 1(a^2 + 1) > 0$$

$$\Rightarrow (3a - 1)(a^2 + 1) > 0$$

$$\therefore a > \frac{1}{3} \Rightarrow a \in \left(\frac{1}{3}, 0\right)$$

$$f(1) < 0 \Rightarrow 1 + 3a - a^2 - 3a^3 < 0$$

$$\Rightarrow 3a^3 + a^2 - 3a - 1 > 0$$

$$\Rightarrow 3a(a^2 - 1) + (a^2 - 1) > 0.$$



$$\Rightarrow (a-1)(a+1)(3a+1) > 0$$

9. Intersection of line when both the planes are same

$$\frac{3}{3\beta^2 + 6(1-2\alpha) + 3} = \frac{-6}{6\alpha^2 + 6(1-2\beta) + 6}$$

$$\Rightarrow 2(\beta-1)^2 + 3(\alpha-2)^2 = 0 \Rightarrow \alpha = 2, \beta = 1$$

$\therefore (\alpha, \beta, 1)$ lies on the planes

$$2x + y + z = 4 \quad (A)$$

$$x + y - 1 = 2 \quad (B) \text{ and } x - 2y = 0 \quad (C)$$

10. $r_1 = 2r_2 = 3r_3$

$$\Rightarrow \frac{r_1}{6} = \frac{r_2}{3} = \frac{r_3}{2} \Rightarrow \frac{\tan\left(\frac{A}{2}\right)}{6} = \frac{\tan\left(\frac{B}{2}\right)}{3} = \frac{\tan\left(\frac{C}{2}\right)}{2}$$

$$\Rightarrow \frac{\tan \frac{A}{2}}{6} = \frac{\tan \frac{B}{2}}{3} = \frac{\tan \frac{C}{2}}{2}$$

$$= \frac{\sqrt{\tan \frac{A}{2} \cdot \tan \frac{B}{2} + \tan \frac{B}{2} \cdot \tan \frac{C}{2} + \tan \frac{C}{2} \cdot \tan \frac{A}{2}}}{\sqrt{6 \times 3 + 3 \times 2 + 2 \times 6}} = \frac{1}{6}$$

$$\therefore \tan \frac{A}{2} = 1, \tan \frac{B}{2} = \frac{1}{2} \text{ \& } \tan \frac{C}{2} = \frac{1}{3}$$

$$\therefore \angle A = \frac{\pi}{2}, \sin B = \frac{2 \cdot \frac{1}{2}}{1 + \left(\frac{1}{2}\right)^2} = \frac{4}{5}$$

$$\sin C = \frac{2 \cdot \frac{1}{3}}{1 + \left(\frac{1}{3}\right)^2} = \frac{3}{5}$$



$$\therefore \cos B = \frac{3}{5} \text{ and } \cos C = \frac{4}{5}$$

$$\therefore (\sin B + \sin C)(\cos B + \cos C) = \left(\frac{3+4}{5}\right)\left(\frac{4+3}{5}\right) = \frac{196}{100} = 1.96$$

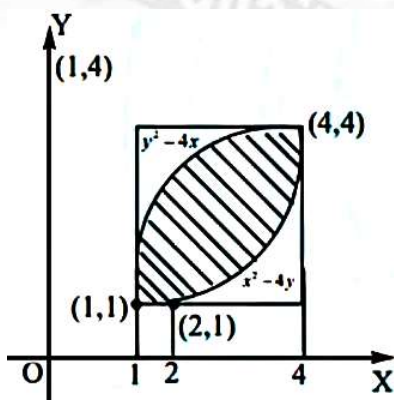
$$11. \quad [\sin x + \cos x] = \left[\sqrt{2} \sin \left(x + \frac{\pi}{4} \right) \right] = \begin{cases} 1 & 0 \leq x \leq \frac{\pi}{2} \\ 0 & \frac{\pi}{2} \leq x \leq \frac{3\pi}{4} \\ -1 & \frac{3\pi}{4} < x < \pi \\ -2 & \pi < x \leq \frac{5\pi}{4} \\ -1 & \frac{5\pi}{4} < x \leq \frac{3\pi}{2} \\ 0 & \frac{3\pi}{2} < x \leq 2\pi \end{cases}$$

$$\begin{aligned} I &= 20 \left[\int_0^{\pi/2} 1 dx + \int_{\pi/2}^{3\pi/4} 0 dx + \int_{3\pi/4}^{\pi} -1 dx + \int_{\pi}^{5\pi/4} -2 dx + \int_{5\pi/4}^{3\pi/2} -1 dx + \int_{3\pi/2}^{2\pi} 0 dx \right] \\ &= 20 \times \left[\frac{\pi}{2} - \pi + \frac{3\pi}{4} - 2 \times \frac{3\pi}{4} + 2\pi - \frac{7\pi}{4} + \frac{3\pi}{2} \right] = 20 \times -\pi = -20\pi \end{aligned}$$

12. The two curves intersect at (4,4) which is a vertex of the given square. Therefore

Required area (Shaded portion) = $\int_1^4 2\sqrt{x}$

$$-\int_1^4 \frac{x^2}{4} dx - (2-1) \times 1 = 2 \times \frac{2}{3} \left[x^{3/2} \right]_1^4 - \frac{1}{12} \left[x^3 \right]_2^4 - 1$$





$$= \frac{4}{3} \times [8-1] - \frac{1}{12} (64-8) - 1 = \frac{28}{3} - \frac{56}{12} - 1 = \frac{112-56-12}{12} = \frac{44}{12} = \frac{11}{3}$$

13. $P : \operatorname{Re} \left(\sum_{k=1}^5 \lambda^k \right)$

- Since $\lambda = \alpha^6$ and $\gcd(6, 11) = 1$, the powers $\lambda, \lambda^2, \dots, \lambda^{10}$ are simply a permutation of the roots $\alpha, \alpha^2, \dots, \alpha^{10}$.

- The sum of all non-trivial 11th roots of unity is -1 . So, $\sum_{k=1}^{10} \lambda^k = -1$.

- Let $S = \sum_{k=1}^5 \lambda^k$. Then the conjugate sum $\bar{S} = \sum_{k=1}^5 \bar{\lambda}^k = \sum_{k=1}^5 \lambda^{-k} = \sum_{k=1}^5 \lambda^{11-k} = \sum_{j=6}^{10} \lambda^j$.

- Since $S + \bar{S} = 2 \operatorname{Re}(S)$, we have $2 \operatorname{Re}(S) = -1 \Rightarrow \operatorname{Re}(S) = -\frac{1}{2}$.

- P - 3

Q: $\prod_{k=1}^{10} (\mu - \beta^k)$

- Since $\beta = \alpha^2$ and $\gcd(2, 11) = 1$, the terms β^k iterate through all roots $\alpha, \dots, \alpha^{10}$.

- Value $= \frac{\mu^{11} - 1}{\mu - 1}$.

- Since $\mu = \alpha^7, \mu^{11} = (\alpha^{11})^7 = 1^7 = 1$.

- The numerator becomes $1 - 1 = 0$.

- Q-2

R: $\prod_{k=1}^{10} (i - \beta^k)$

- Similarly, this product is the polynomial $\frac{z^{11} - 1}{z - 1}$ evaluated at $z = i$.

- Value $= \frac{i^{11} - 1}{i - 1} = \frac{i^3 - 1}{i - 1} = \frac{-i - 1}{i - 1}$.

- Simplify: $\frac{-(1+i)}{-(1-i)} = \frac{1+i}{1-i} \cdot \frac{1+i}{1+i} = \frac{(1+i)^2}{1^2 - i^2} = \frac{2i}{2} = i$.

- R - 4

S: $50(m+n)$

- Combining the results: $Z = \left(-\frac{1}{2}\right) + (0) + (i) = -\frac{1}{2} + i$.

- Here, $n = -\frac{1}{2}$ and $m = 1$.

- Value $= 50(m+n) = 50(1-0.5) = 50(0.5) = 25$.



- S - 1

14. Domain: $x \in (\sqrt{2}, 2] \cup \{0\}$

$x = 0$ is not the solution

$\forall x \in (\sqrt{2}, 2], f(x) = \sin^{-1}(x-1)^2 + \cos^{-1}\left(\frac{1}{x^2-1}\right)$ and $g(x) = \tan^{-1}x$ are strictly

increasing functions, where $f(\sqrt{2}) < g(\sqrt{2})$ and $f(2) > g(2)$ so 1 solution

15. $\lim_{n \rightarrow \infty} \sum \frac{k^3 + 6k^2 + 11k + 6}{(k+3)!} - \frac{1}{(k+3)!}$

$$\sum \frac{(k+1)(k+2)(k+3)}{(k+3)!} - \frac{1}{(k+3)!}$$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k!} - \frac{1}{k+3!}$$

$$\frac{1}{1 + \frac{1}{c^4} + \frac{1}{c^2}} - \frac{1}{1/6} + \frac{1}{13} - \frac{1}{16} + \frac{1}{14} - \frac{1}{17} + \frac{1}{15}$$

$$\frac{1}{1 + \frac{1}{12}} - \infty$$

16. Equation of asymptotes $\sqrt{3}x - y + 5 = 0$

$$y = \sqrt{3}x + 5$$

$$y = -\sqrt{3}x + 1$$

Angle between Asymptotes $= 60^\circ = \theta$ eccentricity $\Rightarrow \sec\left(\frac{\theta}{2}\right) = \sec 30^\circ = \frac{2}{\sqrt{3}}$

17. $3\cot^2 x + 8\cot x + 3 = 0$

$$\cot x = \frac{-8 \pm \sqrt{64 - 4 \cdot 3 \cdot 3}}{2 \cdot 3}$$



$$\cot x = \frac{-8 \pm \sqrt{64 - 36}}{6}$$

$$\cot x = \frac{-8 \pm \sqrt{28}}{6}$$

$$\cot x = \frac{-8 \pm 2\sqrt{7}}{2.3} = \frac{-4 \pm \sqrt{7}}{3}$$

18. $\vec{a} = 2i + 3j$ $\vec{b} = xi + yj$

$$\vec{a} \cdot \vec{b} = 0$$

Given $\frac{\vec{c} \cdot \vec{a}}{|\vec{a}|} = \frac{5}{\sqrt{13}}$ & $\frac{\vec{c} \cdot \vec{b}}{|\vec{b}|} = \frac{3}{\sqrt{13}}$

$$\hat{a} = \frac{2}{\sqrt{13}}\hat{i} + \frac{3}{\sqrt{13}}\hat{j} \quad \text{since } \vec{b} \perp \vec{a}$$

$$\text{So } \vec{b} = 3\hat{i} - 2\hat{j}$$

$$\hat{b} = 3\hat{i} - \frac{2}{\sqrt{13}}\hat{j}$$

now

$$\vec{c} = \frac{5}{\sqrt{13}}\hat{a} + \frac{3}{\sqrt{13}}\hat{b}$$

put value of $\hat{b}$ & $\hat{a}$ find $\vec{c}$.

19. $f(x) = f(2 - x)$
 $x = 2 - x$

$$f(x - 1) = f(x + 1)$$

graph is symmetric about line $x = 1$

$$\text{Also } f(x) = f(2 - x)$$

If one root is α other will be $2 - \alpha, 2 - \alpha, \beta, 2 - \beta, 1, 1$

Equal roots

Now sum of roots = 6



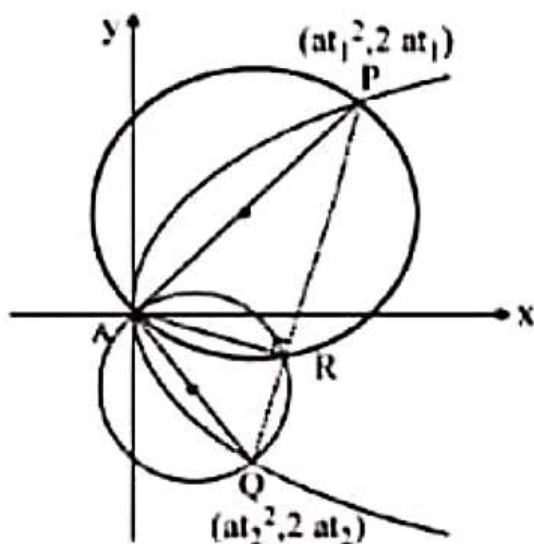
20. $f(x)$ is strictly increasing since the integrand $\frac{t}{1+t^4} > 0$ for $t > 0$.

Its maximum value (as $\rightarrow \infty$) is

$$\int_0^{\infty} \frac{t}{1+t^4} dt = \frac{\pi}{4\sqrt{2}}$$

- The given RHS $\frac{\pi}{8\sqrt{2}}$ is half of the maximum, so the equation has exactly one positive real solution by monotonicity.
- Thus A is true and R is true, and R explains A.

21.



Slope of tangent at P is $\frac{1}{11}$

and at Q $\frac{1}{12}$

$\Rightarrow \cot \alpha = t_1$ and $\cot \beta = t_2$

Slope of PQ $\frac{2}{t_1 + t_2}$

$\Rightarrow$ Slope of AR is $\left(-\frac{t_1 + t_2}{2} \right) = \tan \gamma$

Note:- angle in a semicircle is always 90° .



$$\Rightarrow \tan \gamma = -\frac{1}{2}(\cot \alpha + \cot \beta)$$

$$\Rightarrow \cot \alpha + \cot \beta - 2 \tan \gamma$$

$$\Rightarrow |(\cot \alpha + \cot \beta) \cot \gamma| = 2$$

22. E = 2 balls drawn, none of them is white.

E_1 = both balls red

E_2 = both balls black

E_3 = one red, one black

$$P(E_1 / E) = \frac{P(E_1 / E_1) \cdot P(E_1)}{P(E_1 / E_1) \cdot P(E_1) + P(E / E_2) \cdot P(E_2) + P(E_1 / E_3) \cdot P(E_3)}$$

$$= \frac{\frac{{}^3C_2}{{}^7C_2}}{\frac{{}^3C_2}{{}^7C_2} + \frac{{}^2C_2}{{}^7C_2} + \frac{{}^3C_2 \cdot {}^2C_1}{{}^7C_2}} = \frac{{}^3C_2}{{}^3C_2 + {}^2C_2 + {}^3C_1 \cdot {}^2C_1}$$

$$= \frac{3}{3+1+3 \cdot 2} = \frac{3}{10}$$

23. L.H.S.

$$16 + \sin^2 x + \cos^{11} x - (1 + \tan^2 y) - 9(1 + \cot^2 y)$$

$$6 + \sin^{12} x + \cos^{11} x - (\tan^2 y + 9 \cot^2 y)$$

I

II

Part I ≤ 1 & part II ≤ -6 (using A.M. $\geq$ G.M.) so L.H.S. ≤ 1

$$\text{R.H.S. } \sec^8 z - 2 \sec z + \log_2(5 - \cos^2 x) \geq \sec^2 z - 2 \sec z + \log_2(5 - \cos^2 x) \geq 1$$

Which is possible only when $\cos x = 1$

$$\Rightarrow x = 0, 2\pi$$

$$\text{and } \tan^2 y = 3 \Rightarrow y = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$\text{and } \sec z = 1 \Rightarrow z = 0, 2\pi$$



$$24. \quad [x]\{x\} = 1 \Rightarrow x \in \left\{ n + \frac{1}{n} : n \in I, n \geq 2 \right\}$$

$$[x]\{x\} = 1 \Rightarrow x \in \left\{ n + \frac{1}{|n|} : n \in I, |n| \geq 2 \right\}$$

$$|\alpha|_{(\min)} = 5/2$$

$$|\beta|_{(\min)} = 3/2$$

$$25. \quad \lim_{x \rightarrow 0} \frac{2 \sin \frac{3x}{2} \cdot \cos \left(\frac{3x-2a}{2} \right) - 3 \left(2 \sin \frac{x}{2} \cdot \cos \frac{3x+2a}{2} \right)}{x^3}$$

$$= \lim_{x \rightarrow 0} 2 \cos \left(\frac{3x+2a}{2} \right) \left(\frac{-4 \left(\frac{x}{2} \right)^3}{x^3} \right)$$

$$= -\cos a$$



PHYSICS

26. Position of student = vt (1)

Position of Bus = $16 + \frac{1}{2} \times 2t^2$ (2)

From 1 and 2

$$t^2 + 16 - vt = 0$$

So $v^2 - 64 \geq 0$
 $v \geq 8 \text{ m/s}$

27. We have (V = applied voltage)

$$\frac{V^2}{R_1} \times 4 = \frac{V^2}{R_2} \times t = \frac{V^2}{R_1 R_2} (R_1 + R_2) \times 3$$

Solving we get $t = 12$ minutes

28. Since each body in equilibrium

So $F_B = W = Mg$

29. For 1st pin $\frac{1}{V} - \frac{1}{(-9)} = \frac{1}{f}$ (1)

For 2nd pin $\frac{1}{-V} - \frac{1}{(-18)} = \frac{1}{f}$ (2)

For one pin image would be real and for other virtual from 1 and 2

$f = +12 \text{ cm}$ (convex)

30. Conceptual

31. For positron emission

$$Q = \left\{ M_{Al} - (M_{Mg} + 2m_e) \right\} c^2$$

$m_e c^2 = 0.511 \text{ MeV}$ (given)

$Q = 3.256 \text{ MeV}$

32. Decrease in surface energy = ΔkE

$$\Delta U_S = \frac{1}{2} mv^2 = mgh$$

So $h \propto \frac{\Delta U_S}{m} \Rightarrow h \propto \frac{r^2}{r^3}$

33. MI of the triangle proportional to square of the side and mass, since mass is also proportional to side so MI of the removed triangle reduces by factor of 16.

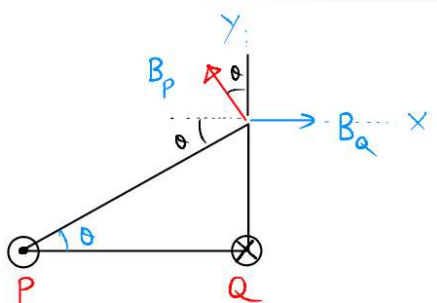
Hence MI of the remaining plate will be $I_0 - \frac{I_0}{16} = \frac{15I_0}{16}$



34. Conceptual

35. In the given situation breakdown does not occur and it will same current flows through R_S and R_L

36. $\vec{B}_Q = \frac{\mu_0}{4\pi} \frac{21}{3} \hat{i}$



$$\vec{B}_P = \frac{\mu_0}{4\pi} 21 \{-\sin\theta \hat{i} + \cos\theta \hat{j}\}$$

37. $\frac{Q}{A \in_0 K} \leq \text{Dielectric strength}$

$$\frac{CV}{A \in_0 K} \leq E$$

$$\Rightarrow A \geq \frac{CV}{\in_0 K E}$$

$$\frac{\pi d^2}{4} \geq \frac{cV}{\in_0 K E}$$

$$\frac{\pi d^2}{4} \geq \frac{6.4}{8.85} \text{ (Solving)}$$

$$\Rightarrow d^2 \geq 0.92$$

$$d \geq 96 \text{ cm}$$

(Option are far spaced so smart approximates can be done)

38. Magnetic field points into the plane of drawing, when current in the straight conductor increases then in accordance with Lenz law, current is induced in ACW sense and loop experiences repulsive force, similar argument leads to the correct choice.

39. Apply conservation of linear momentum for the system along horizontal direction.

40. Under the action of internal forces, velocity of centre of mass does not change so Assertion is not correct. Reason is a fact.



41. Conceptual

42. $\boxed{Q = \Delta U + W}$ (1)

$$\Delta U = \frac{3R\Delta T}{2}, W = R\Delta T$$

$$Q = 8.31 \times 60 \times 60$$

From (1) and given data

$$\Delta T = 1440$$

$$\Rightarrow T_2 = 1740 \text{ K}$$

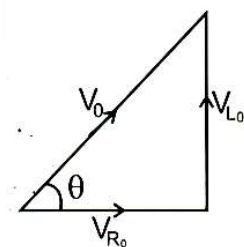
$$W = 1440 R \text{ and } Q = 2160 R$$

$$\frac{V_2}{V_1} = 5.8$$

43. Due to elastic nature of collision, relative velocity does not change

44.

$$\vec{i}_{R_0} = \vec{i}_{L_0} = \vec{i}_0$$



$$V_R = iR$$

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\omega = 1 \tan \phi = \frac{X_L}{R} = 1 \Rightarrow \phi = \frac{\pi}{4}$$

$$V = -V_0 \cos\left(t + \frac{\pi}{4}\right)$$

$$= V_0 \sin\left(\frac{3\pi}{2} + t + \frac{\pi}{4}\right)$$

$$i = \frac{V_0}{2} \sin\left(t + \frac{3\pi}{2} + \frac{\pi}{4} - \phi\right) = \frac{V_0}{\sqrt{2}} \sin\left(\frac{3\pi}{2} + t\right)$$

$$i = -\frac{V_0}{\sqrt{2}} \cos t \Rightarrow V_R = iR = -\frac{V_0}{\sqrt{2}} \cos t$$

45. When screen is moved towards, distance D (usual meaning) decreases and so fringe width decreases, but angular fringe width remains same. In case of white light, we get superposition of interference pattern of individual colours and hence due to overlapping



we observe Maxima at the centre surrounded by coloured fringes. When YDSE is immersed in a liquid, wavelength decreases and fringe width decreases

$$46. \quad [P^x E^y C^z] = [M^0 L^0 T^0] \quad \dots (1)$$

$$[\text{Radiation Pressure}] = \left[\frac{(M L T^{-2})}{L^2} \right]$$

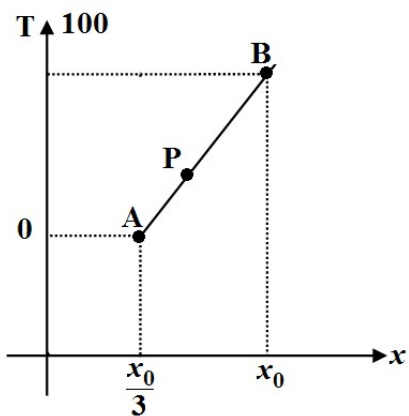
$$\frac{[E]}{\text{Intensity}} = \frac{[M L^2 T^{-2}]}{L^2 T}$$

From (1) we get $x + y = 0$

$$x - z = 0$$

$$\text{and } -2x - 3y - z = 0$$

47.



$$P\left(\frac{x_0}{3}, T\right)$$

$$(\text{Slope})_{AB} = (\text{Slope})_{AP}$$

$$\frac{100 - 0}{\left(\frac{2x_0}{3}\right)} = \frac{T - 0}{\frac{x_0}{6}}$$

$$\Rightarrow T = 25^\circ \text{C}$$

48. Voltage across points A and D will be 62 volts, across AC it will be 40 volts.

$$49. \quad \frac{E\left(\frac{R}{2}\right)}{E_R} = 0.125$$

$$\text{Let } \rho = k.r^n$$



$$E_R \propto \frac{\int_0^R \rho \cdot 4\pi r^2 dr}{R^2} \quad \dots (1)$$

From (1) and (2), we get $n = 2$

$$E\left(\frac{R}{2}\right) \propto \frac{\frac{R}{2} \int_0^{\frac{R}{2}} \rho \cdot 4\pi r^2 dr}{\left(\frac{R}{2}\right)^2}$$

50.



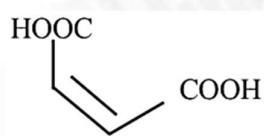
$$V = \sqrt{\frac{T}{\mu}} = 10 \text{ m/s}$$

$$f = \frac{V}{\lambda} \Rightarrow \lambda = 0.1 \text{ m}$$

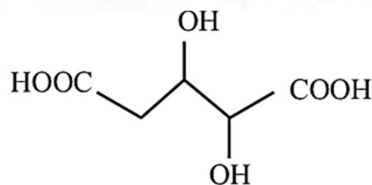
So, separation between consecutive nodes $= \frac{\lambda}{2} = 5 \text{ cm}$

CHEMISTRY

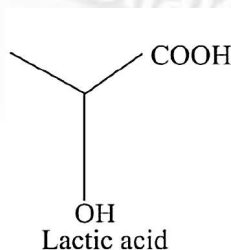
51.



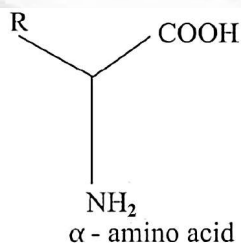
maleic acid



Tartaric acid



Lactic acid

 α - amino acid

Maleic acid has no chiral centre, so optically inactive.

52. For zero order reaction, $x = kt$



$$\therefore \frac{a}{2} = k \times t_1 / 2, \text{ i.e. } t_1 / 2 = \frac{a}{2k} \quad \dots(i)$$

$$\text{For first order reaction, } 1/2 = \frac{\log_e 2}{k} \quad \dots(ii)$$

$$\text{From (i) and (ii), } \frac{a}{2k} = \frac{\log_e 2}{k}$$

$$a = \log_6 4M$$

Hence, (A) is the correct answer.

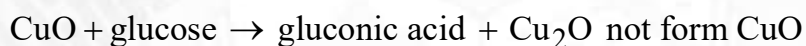
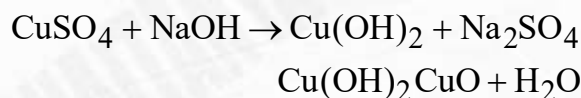
53. EA (Cl) > EA(F)

Down the group radius increases

Correct order: B < C < O < N

For isoelectronic species, size increase as $\frac{Z}{e}$ decreases.

54. Because



55. $\text{HA} + \text{BOH} \rightleftharpoons \text{BA} + \text{H}_2\text{O}$

$$t = 0 \quad a \quad 12 \times 0.1 \quad 0 \quad 0$$

$$t = \text{eq} (a - 1.2) \quad 0 \quad 1.2 \quad 1.2$$

Meq of HA = Meq of BOH at end point

$$A = 26.6 \times 0.1 = 2.66$$

$$\therefore \text{Meq of HA left} = 2.66 - 1.2$$

$$\text{Now, } \text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]_{\text{left}}}$$

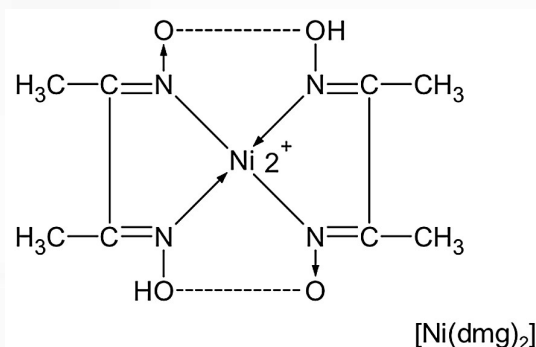


$$5 = -\log K_0 + \log \frac{1.2}{(2.66 - 1.2)}$$

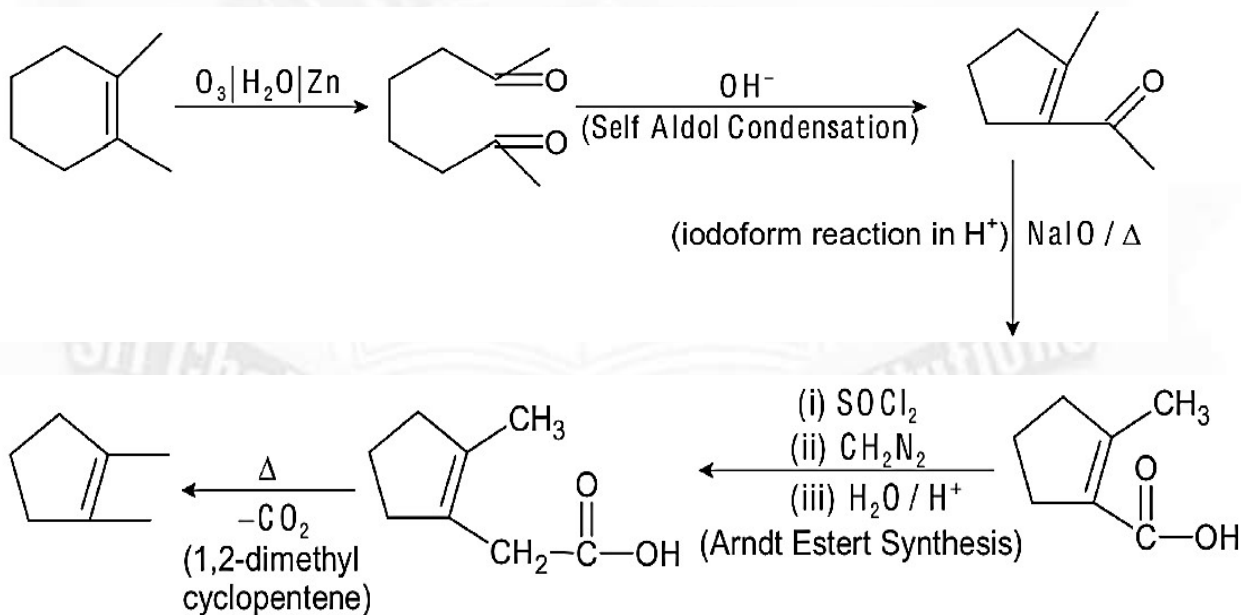
$$K_a = 8.219 \times 10^{-6}$$

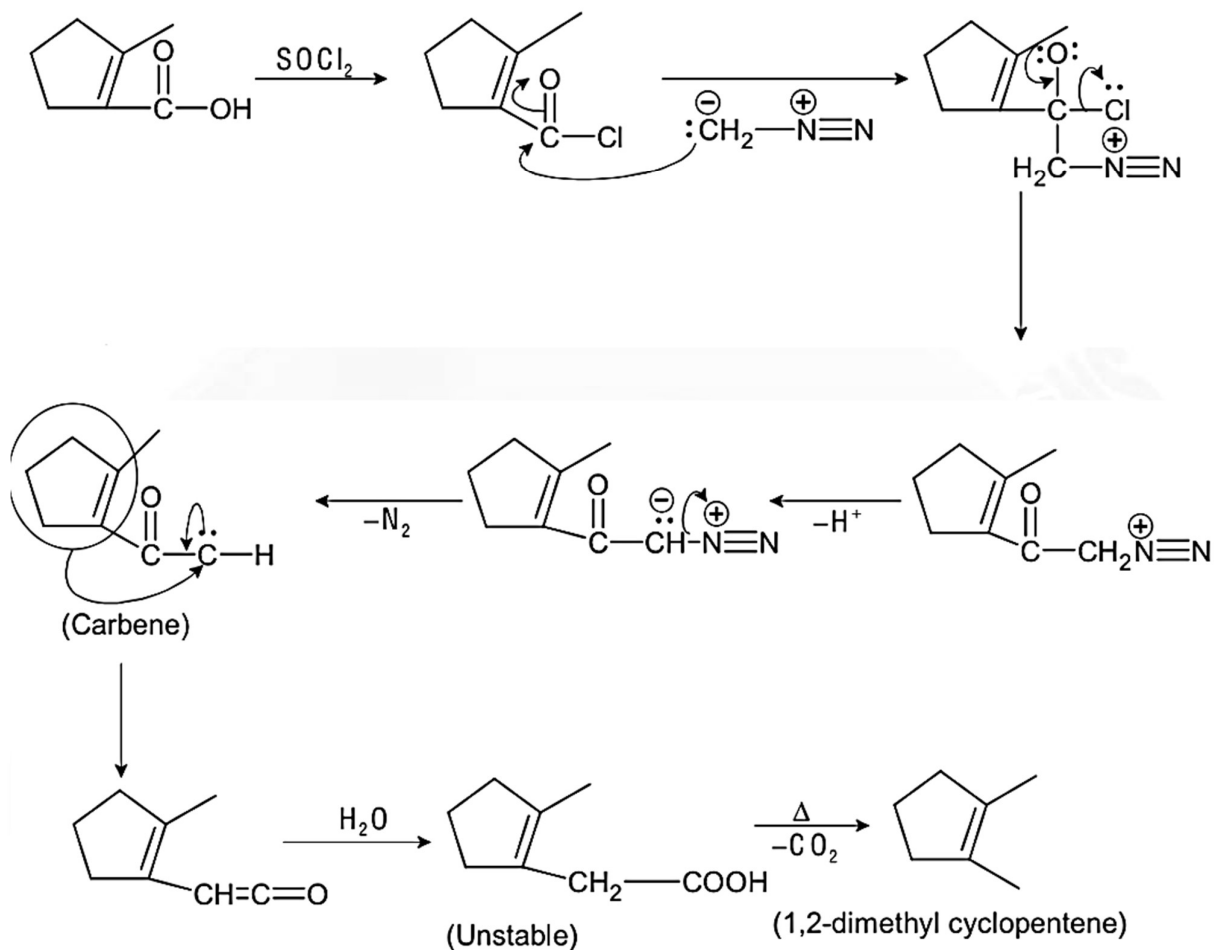
56. Energy of n th state in H-atom is same as energy of 3 n th state in Li^{++} .

57.



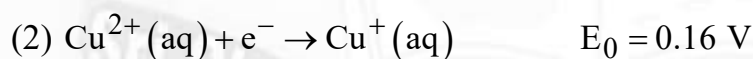
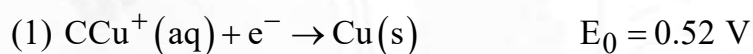
58.



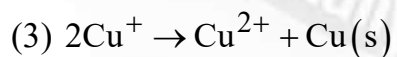


59.

Given,



From (1) and (2), we get



From equation (1), we get

$$\Delta G_1 = -nFE_0 = -1 \times F(0.16 \text{ V}) = -0.16FV$$

From equation (2), we get



$$\Delta G_2 = -nFE_0 = -1 \times F \times 0.52 \text{ V} = -0.52F \text{ V}$$

$$\therefore \Delta G = -0.52FV + 0.16FV$$

$$\Rightarrow \Delta G = -0.36FV = -0.36 \times 96500 \text{ kJ}$$

$$\Rightarrow \Delta G = -34.740 \text{ kJ}$$

Correct option is (a)

60. $1s^2 2s^2 \rightarrow \text{Be}$

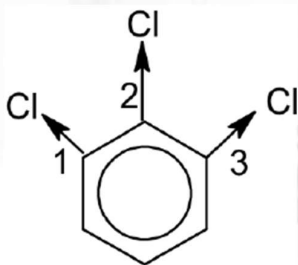
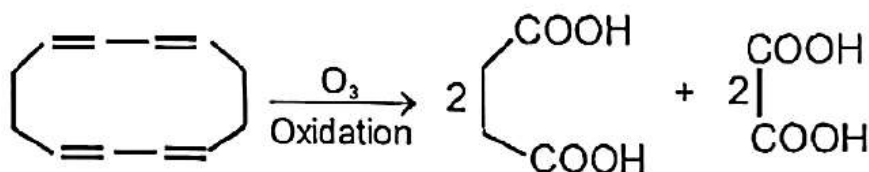
$$1s^2 2s^2 2p^6 \rightarrow \text{Ne}$$

$$1s^2 2s^2 2p^3 \rightarrow \text{N}$$

$$1s^2 2s^2 2p^5 \rightarrow \text{F}$$

$$\text{Ne} > \text{E} > \text{N} > \text{B} \quad (\text{order of I.E.})$$

61.



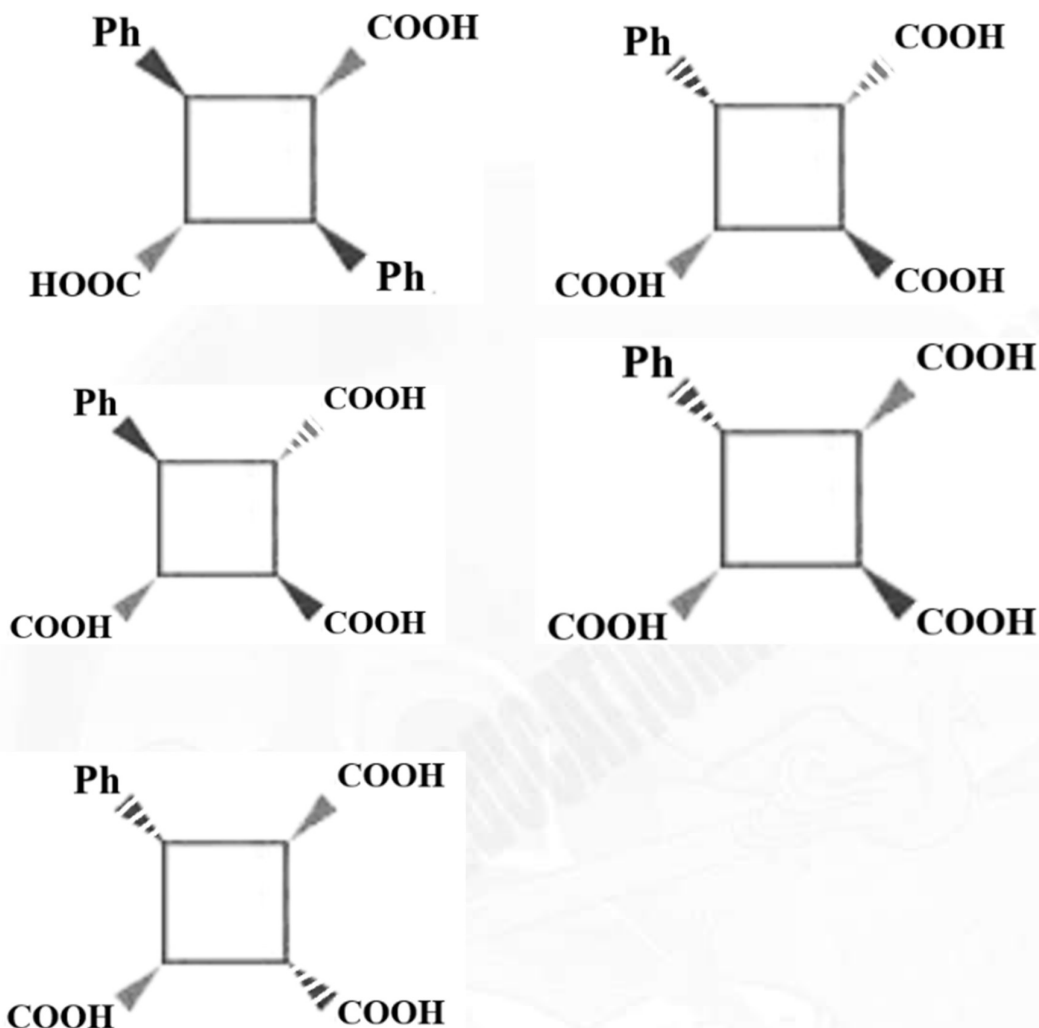
62. , resultant bond moment of $C_1 - \text{Cl}$ and $C_3 - \text{Cl}$ bonds adds up to the bond moment of $C_2 - \text{Cl}$. Hence, it has maximum value.

63. $\pi = CRT = \frac{n}{v} RT = \frac{W}{M} \times \frac{1}{V} \times RT$

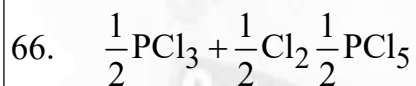
$$7.65 = \frac{W}{180} \times \frac{1}{1} \times 0.0821 \times 310 \quad W = 54.18 \text{ g}$$



64.



65. MOT

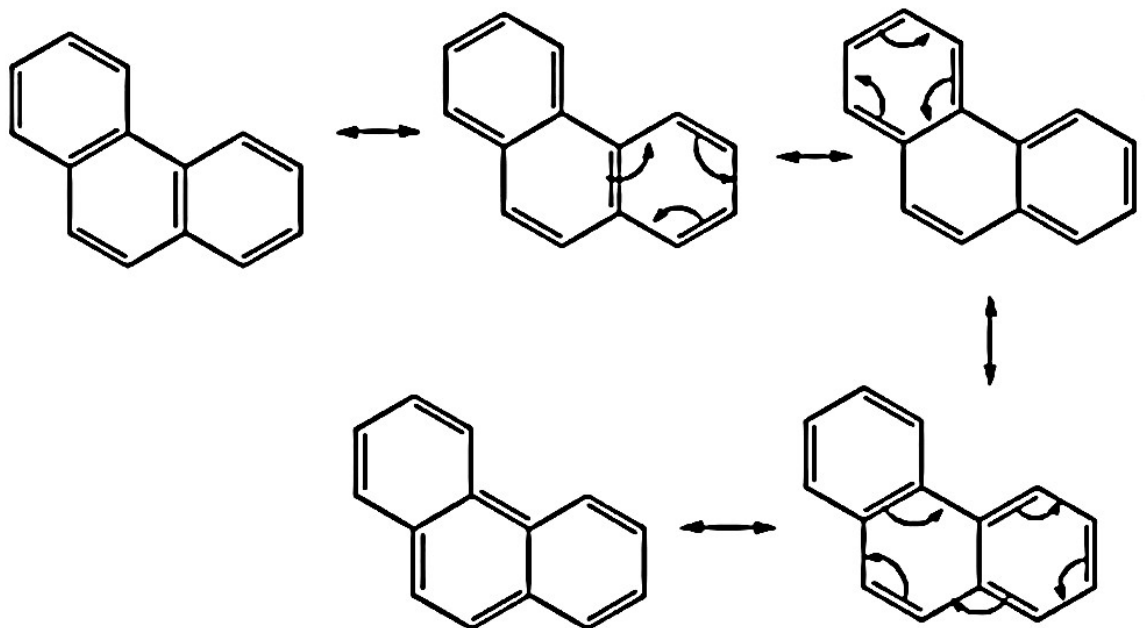


at equilibrium 2 mole 2 mole 2 mole
 4 atm 4 atm 4atm

$$K_P = \left(\frac{P_{\text{PCl}_5}}{P_{\text{PCl}_3} \times P_{\text{Cl}_2}} \right)^{1/2} = \left(\frac{4}{4 \times 4} \right)^{1/2} = 2 \text{ atm}^{-1/2}$$



67.



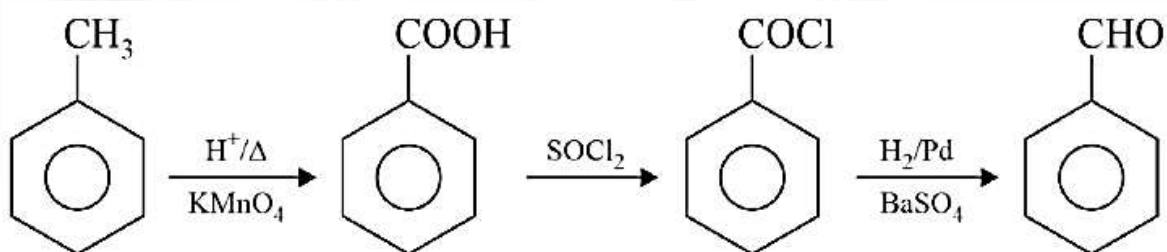
68. 1 mole of A will form 2 moles of B and C after completion of reaction when 75%. A converted into B and C then total no. of moles

69. $\text{Yb}^{2+} (f^{14})$ is a strong reducing agent as it converts to common +3 oxidation state.

$\text{Tb}^{+4} (\text{TbO}_2)$ is an oxidizing agent and it converts to common +3 oxidation state.

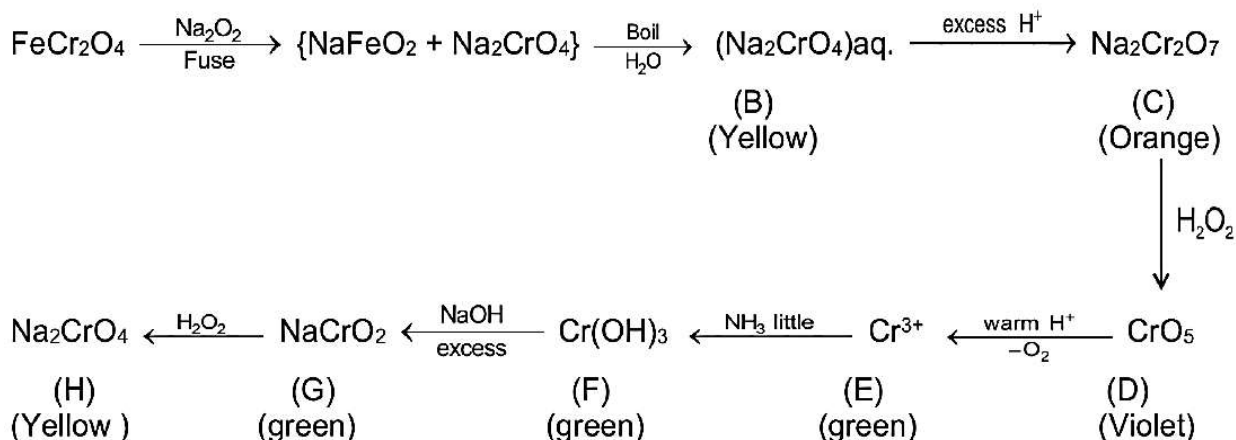
So, both statements are correct.

70.





71.



72. Given that $\frac{1}{[\text{Reactant}]}$ vs time plot is a straight line, the reaction is of second order and intercept gives the initial concentration (intercept = $1/a$) and slope = k .

Therefore, $a = \frac{1}{40 \times 10^2} = 0.25 \times 10^{-2} \text{ mol}^{-1} \text{ dm}^3$

$$k = 4.0 = \text{mol dm}^{-3} \text{ s}^{-1}$$

Now half-life period for second order reaction is given as

$$t_{1/2} = \frac{1}{ka} = \frac{1}{4 \text{ mol dm}^{-3} \text{ s}^{-1} \times 0.25 \times 10^{-2} \text{ mol}^{-1} \text{ dm}^3} = 100 \text{ sec}$$

73. We have $C_p(T) = 20.0 + 0.01T \left(\text{J K}^{-1} \text{ mol}^{-1} \right)$. For an ideal gas,

$$\Delta H = \int_{T_0}^{T_1} C_p(T) dT$$

So for $T_1 = 300 \text{ K}$ and $T_2 = 400 \text{ K}$:

$$\Delta H = \int_{300}^{400} (20.0 + 0.01T) dT = 20.0(400 - 300) + \frac{0.01}{2}(400^2 - 300^2)$$

Calculate:

$$20.0 \times 100 = 2000 \text{ J mol}^{-1}$$

$$\frac{0.01}{2} = 0.005, 400^2 - 300^2 = 160000 - 90000 = 70000,$$



$$0.005 \times 70000 = 350 \text{ J mol}^{-1}$$

Therefore

$$\Delta H = 2000 + 350 = 2350 \text{ J mol}^{-1}$$

Answer: $\Delta H = 2350 \text{ J mol}^{-1}$ (increase in enthalpy).

74. $K_{sp}(\text{Pb}(\text{OH})_2) = 4 \times 10^{-12}$

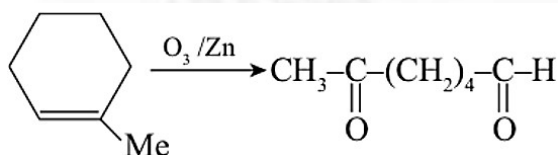
Solubility in buffer $K_{sp} = 4 \times 10^{-12} = [\text{Pb}^{+2}][\text{OH}^-]^2$

In buffer of pH = 10, $[\text{OH}^-] = 10^{-4} \text{ M}$

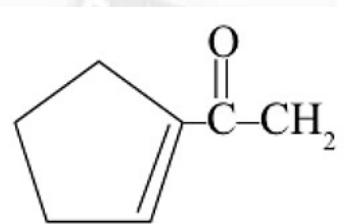
$$[\text{Pb}^{2+}] = \text{solubility in buffer} = \frac{K_{sp}}{[\text{OH}^-]^2} = \frac{4 \times 10^{-12}}{(10^{-4})^2} = 4 \times 10^{-4} \text{ M}$$

$$\text{Ratio} = \frac{4 \times 10^{-4}}{10^{-4}} = 4 \text{ times}$$

75.



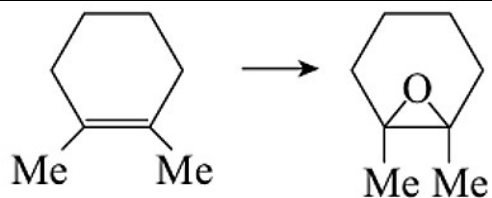
Now α H of ketone is involved in aldol condensation reaction, product is



$$X = 1$$

This is Cross Cannizaro Reaction

$$\therefore Y = 2$$

 $Z = 1$

X	Y	Z
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2	2	1
---	---	---